Table C1. Fully unconstrained random parameters binary logit models with heterogeneity in means and variances of crash injury severity (serious injury/fatal vs. PDO/minor injury), estimated separately for each two-year period: single-motorcycle crashes, Thailand, 2016–2023. Estimated parameter (*t*-statistic).

Variable description	2016–2017	2018–2019	2020–2021	2022–2023
Constant [serious injury / fatal]				
Constant	0.643 (2.58)	0.852 (5.30)	0.438 (2.37)	–0.352 (–1.93)
Vehicle involvement				
Pedestrian involved (1 if yes, 0 otherwise)		–0.623 (–2.04)		
Crash type				
Rear-end collision in the same lane (1 if yes, 0 otherwise)				0.750 (2.14)
Ran off road to the left and hit an object (1 if yes, 0 otherwise)			0.482 (1.76)	
Sideswipe (parallel movement) (1 if yes, 0 otherwise)		–0.634 (–2.07)	0.832 (2.09)	
Ran off road to the left (1 if yes, 0 otherwise)	–0.191 (–1.02)	–1.119 (–7.39)	–0.535 (–3.06)	–0.375 (–2.34)
(Standard deviation of parameter distribution)	2.071 (2.22)			
Ran off road to the right (1 if yes, 0 otherwise)	–0.434 (–2.17)	–0.689 (–3.05)	–0.608 (–2.33)	
Loss of vehicle control (1 if yes, 0 otherwise)		–1.418 (–7.63)	–1.418 (–7.14)	–1.119 (–6.02)
Ran off road on a left bend (1 if yes, 0 otherwise)		–0.711 (–2.92)		
Ran off road on a right bend (1 if yes, 0 otherwise)		–0.899 (–4.04)		
Crash cause				
Speeding (1 if yes, 0 otherwise)	–0.453 (–1.97)			0.377 (2.23)
Cutting in too closely (1 if yes, 0 otherwise)	–0.665 (–2.32)		–0.988 (–3.08)	
Driver fell asleep (1 if yes, 0 otherwise)				1.449 (3.22)
Drunk driving (1 if yes, 0 otherwise)	–0.732 (–2.69)		0.485 (1.90)	
Roadway characteristics				
Crash on the main carriageway (1 if yes, 0 otherwise)	–0.631 (–2.54)			0.654 (2.25)
Crash on a frontage road (1 if yes, 0 otherwise)	–1.371 (–4.32)	–1.060 (–3.54)		
Road under construction (1 if yes, 0 otherwise)		0.928 (2.02)		
Painted median (1 if yes, 0 otherwise)			–0.469 (–1.97)	–0.551 (–2.20)
Raised or earth median (1 if yes, 0 otherwise)	–0.431 (–2.79)	0.126 (0.73)	–0.338 (–2.15)	
(Standard deviation of parameter distribution)		1.705 (3.15)		
Depressed median (1 if yes, 0 otherwise)	0.384 (2.08)	0.280 (1.90)		0.566 (3.10)
Barrier median (1 if yes, 0 otherwise)	–0.519 (–2.14)		–0.584 (–2.52)	–0.490 (–2.63)
Horizontal curve (1 if yes, 0 otherwise)	0.868 (2.17)		0.207 (0.53)	
(Standard deviation of parameter distribution)	2.327 (2.20)		2.490 (1.82)	
Concrete pavement surface (1 if yes, 0 otherwise)		–0.513 (–2.90)		–0.368 (–2.04)
Environmental characteristics				
Raining (1 if yes, 0 otherwise)		0.885 (2.83)		
Dark, no street lighting (1 if yes, 0 otherwise)	1.341 (5.76)	0.743 (3.80)	0.457 (2.35)	
Dark, with street lighting (1 if yes, 0 otherwise)	0.764 (4.60)	0.889 (5.19)	0.351 (2.39)	
Temporal characteristics				

Variable description	2016–2017	2018–2019	2020–2021	2022–2023
Weekend (1 if yes, 0 otherwise)			–0.985 (–2.12)	
(Standard deviation of parameter distribution)			2.178 (3.17)	
Songkran festival period (1 if yes, 0 otherwise)	–0.549 (–3.48)	–0.788 (–5.73)	–0.679 (–3.22)	–0.987 (–3.48)
(Standard deviation of parameter distribution)				2.146 (2.44)
New Year festival period (1 if yes, 0 otherwise)	–0.514 (–3.56)	–0.973 (–7.07)	–0.596 (–3.82)	–1.046 (–3.54)
(Standard deviation of parameter distribution)				2.373 (2.71)
Evening (1 if yes, 0 otherwise)	–0.718 (–4.29)	–0.598 (–3.86)		
Midnight / early morning (1 if yes, 0 otherwise)			0.643 (3.95)	0.944 (6.30)
Peak hours (1 if yes, 0 otherwise)		–0.281 (–1.70)		
Region				
Eastern region (1 if yes, 0 otherwise)	0.961 (3.35)	0.726 (2.91)	0.815 (2.88)	
Northeastern region (1 if yes, 0 otherwise)	0.622 (4.17)	0.632 (4.49)	0.648 (3.80)	1.345 (7.04)
Northern region (1 if yes, 0 otherwise)			–0.516 (–2.88)	
Southern region (1 if yes, 0 otherwise)				0.354 (1.69)
Western region (1 if yes, 0 otherwise)			1.190 (3.69)	0.548 (2.34)
Heterogeneity in the means of random parameters				
Raised or earth median: Rear-end collision in the same lane		–3.398 (–3.18)		
Horizontal curve: Weekend	–1.090 (–2.08)			
Weekend: Speeding			1.250 (2.32)	
Weekend: Raining			–1.763 (–2.30)	
New Year festival period: Eastern region				2.517 (2.62)
Model statistics				
Number of observations	1,874	1,932	1,791	1,728
Number of estimated parameters	22	24	26	20
Random parameters / heterogeneity in means / in variances	2 / 1 / 0	1 / 1 / 0	2 / 2 / 0	2 / 1 / 0
Log-likelihood at zero, LL(0)	–1298.96	–1339.16	–1241.43	–1197.76
Log-likelihood with constant only, LL(C)	–1297.57	–1338.56	–1238.46	–1197.75
Log-likelihood at convergence, LL(β)	–1192.09	–1162.28	–1096.24	–1059.50
McFadden pseudo $\rho^2 = 1 - LL(\beta)/LL(0)$	0.082	0.132	0.117	0.115

Notes. A positive parameter increases the probability of a serious injury/fatal outcome. Reference categories: Central region, daytime, no median, one motorcycle. Each column is a separate model for one two-year period. Each model allows at most two random parameters, two heterogeneity-in-means terms and one heterogeneity-in-variance term, searched with 100 Halton draws at the 90% level and confirmed with 2,000 draws; non-significant parameters were then removed, so all parameters except constants and the means of random parameters are statistically significant at the 90% level. A blank cell means the variable is not in that two-year period's model. Rows with a standard deviation give the mean of the random parameter in the two-year periods where it is random and a fixed parameter in the other two-year periods. A variable used in a heterogeneity function enters only the random parameter shown in that two-year period and is removed from the main utility. Random parameters normally distributed; 2,000 Halton draws.